

Naïve Bayes Classification

DSC 340 Applied Machine Learning

Spring 2025

The Plan

- How do we model language?
- Introduce naïve Bayes classification
- Discuss probabilities along the way

NLP

Natural Language Processing

NLP

- **Natural Language Processing (NLP)**
- The use of computing and artificial intelligence to analyze, understand, and generate human language.
- Applications that involve NLP:
 - Google Translate
 - Spam detectors
 - Customer support chatbots
 - Sentiment analysis of social media posts

Why is NLP difficult?

- **Ambiguity:** Words and phrases often have multiple meanings
- **Context Dependence:** Understanding of language depends on shared knowledge and cultural references
- **Evolving Language:** Idioms, slang, and meaning change over time
- **Data Sparsity:** Rare expressions are a challenge to modelling
- **Lack of Grounding:** Machines don't have real-world experiences

Bag of Words (BoW)

- Counts word occurrences
- Ignores order
- Easy to implement, forms basis for many methods



Simple Example: BoW

- Document 1:
“I love Shakespeare” $\rightarrow$ [I:1, love:1, Shakespeare:1]
- Document 2:
“He loves Shakespeare” $\rightarrow$ [He:1, loves:1, Shakespeare:1]
- Compare vectors $\rightarrow$ measure similarity
- Trained on **corpus**

N-Grams and Markov Chains

- Predict next word from last n words
 - Unigrams ($n=1$)
 - Bigrams ($n=2$)
 - Trigrams ($n=3$)
- Chain of probabilities
- Good for rudimentary text generation

Example: Markov Chain Text

- Predicts the next word based on the last n words
- Uses probabilities derived from text patterns
- Mimics style but lacks meaning or understanding

Clustering Documents

- Grouping texts by word frequency (BoW)
- Example: identifying topics, genres, or authors

Classification

- Similar to clustering but with predefined labels
- Examples: Spam detection, language identification

Applications in Action

- Search engines (pre-PageRank era)
- Basic chatbots (Eliza-like, but with more data)
- Machine translation (statistical, phrase-based)

Limitations of Classical NLP

- Context windows too small (n-grams)
- Lacks deep semantics or “world model”
- Struggles with nuance, idioms

Naïve Bayes Classifier

Naïve Bayes (EMC 7.2)

Naïve Bayes

- Probabilistic classification method
- Based on **Bayes' theorem**
- Assumes features are **independent** of each other given the class label
 - It's why we call it **naïve**.
- Method assumes all **features** are **categorical**
 - What if continuous? Can **discretize** based on range
 - Could think of it as "one-hot encoding" for continuous features
 - Note: Extensions exist for continuous features
- Output is the **predicted class label** and a **probability score**

Applications

- Text classification
- Document retrieval
- Spam filtering (0: HAM, 1: SPAM): used at Centre College
- Fraud detection (e.g., insurance industry)

Bayes' Theorem

EMC 7.2.1

Bayes' Theorem (or Rule or even Law!)

- Named for Rev. Thomas Bayes, English mathematician

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}, \quad P(B) \neq 0$$

- **Conditional probability:** $P(A|B)$
 - Read: "The probability that A is true **given that B is true.**"
 - What is the probability student X made A on MAT 310 test?
 - Suppose the student is a Rhodes' Scholar candidate? Or got D in Calc III?

Types of probabilities (don't mix these up!)

- **Prior probability:** $P(A)$ **marginal**
- **Posterior probability:** $P(A|B)$ **conditional** (depends on context)
- **Joint probability:** $P(A, B)$ both A and B true (at same time)
- Convention for discrete random variables (sometimes followed):
 - Lower case variable: $P(a|b)$ specific case
 - Upper case variable: $P(A|B)$ all values of variables (table)
 - Can mix and match: $P(A|b)$, $P(b|A)$, $P(a,B)$, etc.

Rules you should know

- **Sum rule:** $P(A) = \sum_B P(A, B)$
 - Use summation if B is discrete.
 - Use integration (calculus) if B is continuous.
- **Product rule:** $P(A, B) = P(A|B)P(B)$
- **Bayes' theorem:** $P(A|B) = P(B|A)P(A)/P(B)$

General Form for Classification

$$P(C_k|\mathbf{x}) = \frac{P(x_1, x_2, \dots, x_D|C_k)P(C_k)}{P(x_1, x_2, \dots, x_D)}$$

- **Probability of class C_k given** observed feature vector $\mathbf{x}$
- Reminder: $P(x_1, x_2)$ is the **joint probability**.

Independence

- **Important:** $P(A, B) = P(A)P(B)$ only if A and B are **independent**
 - Notation: $A \perp B$
 - **You are not allowed to just go around multiplying probabilities!**
- **Conditional independence:**
 - A and B are said to be conditionally independent given C iff:
 - $P(A|B, C) = P(A|C)$
 - Notation: $A \perp B \mid C$
 - **Can't assume probabilities are independent conditioned on class.**
- **Naïve Bayes classifier assumes you can!! (hence naïve)**

How the Classifier Works

EMC 7.2.2

Naïve Bayes Classifier

- Assumes that probabilities of features
- are conditionally independent
- of all other features
- given the class label.
- Analogy: two bags: think of data as drawn from one or the other
- Given a particular item, which bag is more likely?

$$P(x_1, x_2, \dots, x_D | C_k) = P(x_1 | C_k) P(x_2 | C_k) \dots P(x_D | C_k) = \prod_{j=1}^D P(x_j | C_k)$$

Naïve

- Simplifies computation of $P(x_1, x_2, \dots, x_D | C_k)$
- Also ignores the denominator $P(x_1, x_2, \dots, x_D)$
- Only care about relative probability (which is most likely?)
- Turns out the prediction is directly proportional to true probability
- Also turns out that it is not a very accurate probability!
- But may be good enough (compare perceptron/logistic regression)

Numerical issues

- Probabilities that are exact combinations of features can be very tiny.
- **Numerical underflow** is common.
- Underflow means becomes 0 in CPU, which is very, very, very bad.
- So it is common to work in **log space**.
- Take the log (any base) of all the probabilities
- Then instead of **multiplying**, we can just **add** (slide rule)
- Also note, probabilities don't sum to 1 in this model
- Why does this not matter?

Smoothing

EMC 7.2.3

Smoothing

- Recall that in Bayes' thm. probability cannot be 0
- Bad things happen if probability is 0
- If no data that fits an observation, classifier may assume $p=0$.
- Text model: specific combination of 3 words
- **Laplace smoothing**: also known as **add-one** technique:
 - Assumes there is a minimum of at least 1 of everything, or epsilon
 - Most packages provide this for you: if not, you must implement it

Assessment

EMC 7.2.4

Assessment

- Discuss: How does it compare with other methods we have seen?
- Easy to implement
- Relatively fast
- Performs well in many practical settings (e.g., spam classification)
- Probabilities not to be taken too literally (but same for logistic regr.)
- As George Box said, "All models are wrong, but some are useful."

NB in Python

PDSH